Transportation Research Part E 187 (2024) 103574

Contents lists available at ScienceDirect

Transportation Research Part E

journal homepage: www.elsevier.com/locate/tre

Estimating the potential for optimized curb management to reduce delivery vehicle double parking, traffic congestion and energy consumption

Aaron J. Burns [a], Jeremy J. Michalek [b,*], Constantine Samaras [c]

[a] Department of Engineering and Public Policy, Department of Mechanical Engineering, Carnegie Mellon University, Pittsburgh, PA, 15213, United States of America
[b] Department of Engineering and Public Policy, Department of Mechanical Engineering, Department of Civil and Environmental Engineering, Carnegie Mellon University, Pittsburgh, PA, 15213, United States of America
[c] Department of Civil and Environmental Engineering, Wilton E. Scott Institute for Energy Innovation, Carnegie Mellon University, Pittsburgh, PA, 15213, United States of America

ARTICLE INFO

Dataset link: https://github.com/burnsaar/Parking_Slot_Assignment_Problem-Public

Keywords:
Reservable curbspace
Double parking
Delivery vehicles
Parking slot assignment
Traffic congestion
Fuel consumption

ABSTRACT

We model an optimized curb parking reservation system and characterize conditions under which such a system can reduce delivery vehicle double parking, congestion and energy consumption. We implement an optimization model leveraging integer linear and mixed-integer linear programming parking slot assignment formulations to minimize double parking and build a queuing model to estimate lane obstruction congestion and energy effects. Using delivery data from Aspen, CO and Pittsburgh, PA, we find that, when arrival times are known at the start of the day, a single-space reservation system can eliminate 1–2 min of double parking per hour, on average, increasing 3–4 times when drivers have ±5 minutes of arrival time flexibility. Scenarios involving up to seven co-located spaces show diminishing returns but suggest reservation systems can provide the equivalent of additional parking capacity in some cases. When arrival times are uncertain and buffers are used between reservations, we find that curb reservation systems reduce congestion in high-demand, moderate uncertainty scenarios with short duration reservations but increase congestion in scenarios with high uncertainty and long duration reservations, with estimated annual congestion and emissions externality cost savings ranging from -$100,000 to +$300,000 for a single space. Our results suggest that implementation of curb reservation systems should be targeted to application environments where they can provide net benefits, and our simulations help to characterize key factors for identifying such applications.

1. Introduction

Vehicle deliveries are essential in modern society, but delivery operations also consume energy and create negative externalities, such as congestion, crash risks, air pollution, greenhouse gas (GHG) emissions, and noise (additional details on vehicle externalities in Parry et al. (2007)). This is especially relevant for delivering goods in urban areas where, not only do delivery drivers need to navigate busy streets with large vehicles, but they also need to find a parking space close enough to the target destination to unload or load. Parking spaces, however, can be unavailable at times due to limited supply Jaller et al. (2013), increased e-commerce and

* Corresponding author.
E-mail addresses: aaronbur@andrew.cmu.edu (A.J. Burns), jmichalek@cmu.edu (J.J. Michalek), csamaras@cmu.edu (C. Samaras).

https://doi.org/10.1016/j.tre.2024.103574
Received 19 September 2023; Received in revised form 5 May 2024; Accepted 6 May 2024
Available online 30 May 2024
1366-5545/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

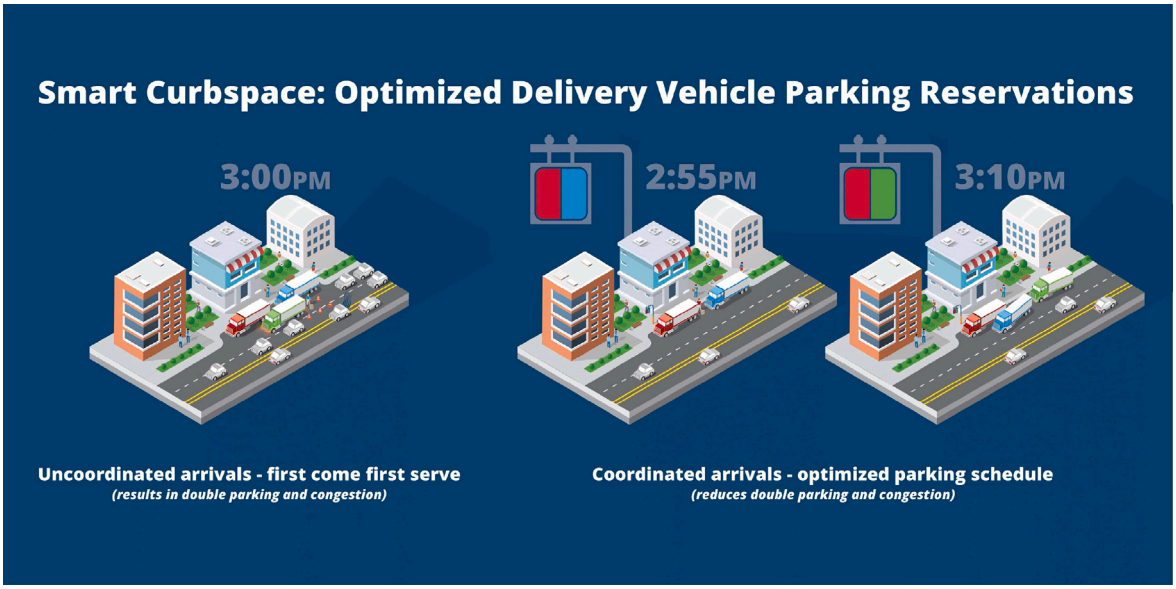


Fig. 1. A smart curbspace system allows for optimized scheduling of delivery vehicle arrivals to parking spaces, potentially reducing the traffic congestion caused by uncoordinated, first-come first-serve delivery vehicles and double parking.

freight demand Dey et al. (2019), Schrank et al. (2019), Spadafora and Rodriguez (2021), competition with other commercial and private vehicles Johansen et al. (2014), Schrank et al. (2019), and unauthorized vehicles in loading zones Coord (2021a), Dey et al. (2019), Girón-Valderrama et al. (2019).

When a parking space is not available, delivery vehicles may exhibit several behaviors including unauthorized parking, "cruising" in search of another parking space, waiting for an open space, or moving to the next delivery Chiara et al. (2021). In this research, we focus on double parking, a form of unauthorized parking where a delivery vehicle parks in a traffic lane next to occupied parking spaces. Through delivery driver surveys, Amaya et al. (2023) found that double parking is a common strategy when parking availability is limited. This is driven by a need to park in close proximity to the delivery address and complete on-time deliveries. Additionally, empirical evidence from New York City indicates a high occurrence of double parking where roughly 350,000 double parking traffic violations were issued to commercial vehicles in Manhattan in 2019 Kim and Wang (2021). We use the term "double-parking" loosely to refer to any unauthorized parking in a lane of traffic (whether the vehicle in the traffic lane is parked next to a legally-parked vehicle or along a section of curb that prohibits parking).

One approach to reducing double parking at dense, congested, and high-demand delivery locations is a "smart curbspace" reservation system that consists of a reservation platform for some or all available curb parking spaces on a city block, as depicted in Fig. 1. These reservable spaces can be placed in various geographically disperse locations, however, in this research we model a set of co-located parking spaces on a single city block. With a reservation system, delivery companies may potentially bid for a parking space for a time and date in the future (we model day-ahead reservations) or place a real-time reservation for a parking space Yang et al. (2019). In a real-time application, reservations may happen automatically via smartphone navigation apps Abidi et al. (2015), Chen et al. (2015), Hakeem et al. (2016), Mladenović et al. (2021).

Evidence for the potential benefits of reservable parking spaces for delivery vehicles is found academic literature and in the commercial startup environment. Castrellon and Sanchez-Diaz (2024) provide insight into the potential benefits of "smart curbspace". They identified in a meta-analysis of recent curbside research that the allocation of urban parking spaces for freight and data sharing, that can enable reservation systems, are top themes. Across the recent literature they found numerous benefits from better parking space management including an average decrease of parking violations and emissions of 41% and a 25% decrease in cruising. Ramirez-Rios et al. (2023) echo the need for allocated dedicated parking spaces for freight and service vehicles in congested urban areas, especially during peak hours based on results from their simulation of parking demand in New York City and other high parking demand US zip codes. The authors also find benefits from a "staggered" delivery strategy that closely resembles an optimally scheduled sequence of vehicles from a curb space reservation system. Additionally, commercial companies, such as Coord and Automotus are implementing versions of managed curbspaces focused on traditional loading zones. Coord aims to enable loading zone reservations for freight vehicles Coord (2021b), and Automotus, Inc. is implementing "smart loading zones" in Pittsburgh, PA to support shorter parking events for local businesses Peterson (2021), Onufer (2020).

Table 1
A summary of relevant studies to improve vehicle parking assignment. Bolded text highlight the specific concepts that when combined define the contributions of our research.

Parking reservation studies and related strategies			
Study	Method[a]	Vehicle(s)[b]	Metrics
Assign vehicles to parking spaces			
Mejri et al. (2016)	SW, H	–	Parking space occupancy, request satisfaction ratio
Mladenović et al. (2021)	SW, O	–	Parking request fulfillment, driving and walking distance
Hakeem et al. (2016)	SW, O	–	Total travel time (driving and walking)
Errousso et al. (2021)	SW, O	**D**, P	Parking request fulfillment, walking distance
Chen et al. (2015)	SW, O	–	Weighted combo of cruising and walking time, revenue
Abidi et al. (2015)	GA	–	**Distance and CO2 emissions saved**
Zhang et al. (2023)	GA	**D**	Loading bay utilization, cost to delivery company
Huang and Hsieh (2022)	H	–	Travel distance, cost of parking space density
Ayala et al. (2011)	GT	–	**Distance traveled, fuel use, CO2 emissions**
Sayarshad et al. (2020)	GT	**–**	**Walking distance, number of cruising vehicles**
Roca-Riu et al. (2015)	**O**	**D**	Number of unscheduled vehicles, total service time
Yang et al. (2019)	**O**	**D**	Parking request fulfillment, total service time
Number, location and configuration of loading bays			
Letnik et al. (2018)	F	**D**	**Fuel use, CO2 emitted, travel time/distance**
Comi et al. (2018)	S	**D**	Time waiting for parking, loading bay utilization
Alho et al. (2018)	H, S	**D**	**Fuel use and traffic delay due to double parking**
Our research contribution			
Our paper	**O**	**D**	**Fuel use and traffic delay due to double parking**

[a] SW: Sliding Window, H: Heuristics, O: Optimization, GA: Genetic Algorithm, GT: Game Theory, F: Fuzzy Clustering, S: Sim.
[b] D: Delivery, P: Private, –: Not Specified.

The objective of our research is to characterize the potential double parking externality effects of a curbspace reservation system under both ideal and more realistic conditions. We address this research objective by implementing an optimal day-ahead curb reservation system for a single location of co-located parking spaces, building on prior work by Roca-Riu et al. (2015) and Yang et al. (2019). Our approach has the advantage of leveraging exact solution methods to guarantee optimality (within given tolerances) so that we can be confident that our solutions represent fair comparisons of the best possible performance of reservation systems against the status quo (instead of the performance of one particular heuristic). However, we are limited to cases that only consider a single set of co-located parking spaces and involve day-ahead reservation requests (with deviations due to uncertainty allowed between reservation time and realized arrival time), which are common in some applications Comi et al. (2018), Roca-Riu et al. (2015), Yang et al. (2019), Zhang et al. (2023)[1] but not others.

2. Literature review

We characterize the effects of a delivery vehicle parking reservation system on double parking and its implications for congestion and energy consumption externalities. We summarize relevant literature on (1) methods used to determine vehicle parking assignments or loading bay design and (2) studies that estimate the implications of double parking.

2.1. Literature on relevant methods

Table 1 summarizes studies that contribute methods to assign vehicles to parking spaces or to design loading bays. We describe each in turn.

Parking slot assignment literature

Methods used for parking slot assignment include sliding time windows, heuristics, and exact optimization approaches, each identified in Table 1 by publication.

Mejri et al. (2016), Mladenović et al. (2021), Hakeem et al. (2016), Errousso et al. (2021), and Chen et al. (2015) each conduct a sliding time window approach where the full parking scenario is iteratively explored and parking decisions are made periodically with new requests and updated parking availability. For each time window, the authors employ an optimization technique to provide

[1] Popular parking reservation apps such as SpotHero and ParkWhiz allow for reserving parking spaces several days prior to arriving.

a parking schedule, with the exception of Mejri et al. (2016) who employs a heuristic. Even though most authors use an exact optimization technique, the limited window of the scenario at each time window does not guarantee an optimal schedule across the entire day.

Abidi et al. (2015), Zhang et al. (2023), and Huang and Hsieh (2022) apply heuristic methods, most often in the form of a genetic algorithm, on the entire scenario of parking requests to search for a near-optimal parking schedule. Abidi et al. (2015) uses a centralized management approach for the parking requests and explores the performance of several versions of a genetic algorithm. Zhang et al. (2023) develops the two-echelon stochastic vehicle routing problem loading bay reservation formulation that is an optimal method, but is not scalable which leads to the incorporation of a genetic algorithm. Huang and Hsieh (2022) also develops an exact mixed-integer linear programming solution to allocate parking spaces to vehicle parking request, but similarly creates a custom algorithm to increase scalability. While heuristics are typically superior in their scalability and ability to solve large complex parking scenarios, heuristics do not guarantee an optimal solution, which can make it difficult to compare across cases. This is because with heuristics it is not known whether one case produces a better result than another because the optimum is truly superior in that case or because the heuristic performed better in that case.

Exact-method optimization formulations have more limited scalability but are well-suited for fair comparisons across cases because optimality can be guaranteed in all cases being compared. Two such formulations align closely with our research question. Roca-Riu et al. (2015) proposed a mixed-integer linear program (MILP) formulation adapted from the vehicle routing problem with time windows to determine the optimal sequence of delivery vehicle parking slot assignments for a set of co-located parking spaces based on delivery request arrival windows. Additional detail on the vehicle routing problem and the time windows variant is available in Irnich et al. (2014) and Desaulniers et al. (2015), respectfully. Yang et al. (2019) builds off of Roca-Riu et al. (2015) with an integer linear programming (ILP) formulation for the parking slot assignment problem using discretized time steps to improve scalability, and they incorporate an auction mechanism.

The parking slot assignment problem also resembles an optimal model developed by Pilz et al. (2022) to assign long-haul delivery vehicles to parking spaces during rest periods and is similar to the berth and quay allocation problem for scheduling the loading and unloading of shipping vessels Roca-Riu et al. (2015). Pilz et al. (2022) seek to better utilize parking resources for delivery vehicles during break periods in between transit periods by modeling a simplistic reservation system and an optimal parking space assignment method with perfect trip information. They find significant improvements with a reservation system when compared with first-come first-serve and similar, but slightly worst performance with the simplistic system compared to the optimized model. Additional details on the berth and quay problem are available in Park and Kim (2003), Golias et al. (2007) and Bierwirth and Meisel (2015).

Further, some studies consider multiple agents and incorporate game theoretic methods to assign parking spaces. Both the "Parking Slot Assignment Games" as described by Ayala et al. (2011) and a non-myopic atomic game by Sayarshad et al. (2020) model how parking spaces can be assigned to maximize a social welfare function given vehicle objectives and attributes such as distance to destination.

Loading zone design literature

A smaller grouping of studies also relevant to improving delivery vehicle parking pertains to defining the best number, location and configuration of loading zones. A subset of these studies that also incorporate energy consumption metrics are included in Table 1. For example, Letnik et al. (2018) determines where to place loading bays based on Fuzzy k-means clustering of desired delivery locations and estimates the change in fuel consumption across configurations. Alho et al. (2018) explores a range of loading bays in both number and single or double parking space configurations with an emphasis on the impact to surrounding vehicle traffic from double parked delivery vehicles including metrics on delay and fuel use. Finally, Comi et al. (2018) developes a process that assesses stakeholder preferences, estimates the required number and location of loading bays, and provides a parking reservation system for delivery drivers, finding that actively managing loading bay assignment can decrease average time spent waiting for parking by 28%.

Metrics

Among the studies in Table 1, Abidi et al. (2015), Ayala et al. (2011), Sayarshad et al. (2020), Letnik et al. (2018), and Alho et al. (2018) all provide metrics on either CO2 emissions avoided, changes in fuel use or the amount of vehicle cruising. For example, Letnik et al. (2018) finds that the optimal location of loading bays can reduce fuel use by 45% per trip with an aggregated estimate of 100,000 liters of fuel saved per year in some cases. Alho et al. (2018) also finds that the number of loading bays in a region is important and can reduce fuel use by up to 68%. Of note, Alho et al. (2018)'s fuel use metric is impacted by double parked delivery vehicles that could not parking in a loading bay and that cause delay and congestion in their micro traffic simulation. Our research implements a similar concept but with a queuing model as described in Section 3.7.

Our contribution

Seeking to ensure fair comparisons of provably optimal solutions across scenarios in our study, we adapt, improve, and apply both the MILP and the ILP formulations of the Parking Slot Assignment Problem from Roca-Riu et al. (2015) and Yang et al. (2019). We find that the MILP formulation is more efficient for some problem instances and the ILP formulation is more efficient for other problem instances, so we leverage both formulations in a hybrid approach that avoids the worst performance algorithm in all cases examined. The optimal parking schedules that result are then used to assess implications for double parking and for traffic congestion and energy externalities, similar to the concept in Alho et al. (2018). The next section describes the methods in the literature to estimate traffic delay induced by a double parked vehicle.

2.2. Literature on effects of double parking

After optimizing the delivery vehicle parking schedule, we need a method to assess the externality impacts of the unassigned vehicles that we assume will double park. The primary externality we intend to study is the change in traffic congestion due to double parked vehicles that can increase vehicle delay and result in unnecessary fuel consumption. Approaches in the literature for estimating changes in traffic flow due to double parking include traffic simulations and queuing models.

Traffic simulations are used to estimate congestion effects as in Alho et al. (2018), Chrysostomou et al. (2019), Delaitre (2009), Kladeftiras and Antoniou (2013). For example, Kladeftiras and Antoniou (2013) characterize a range of congestion impacts of double parking based on outputs from microscopic traffic simulation; Iwan et al. (2018) studied the impact of unloading bays on traffic fluidity via cellular automata simulation, estimating that loading zones can improve average vehicle velocity by 6% to 8%, reducing emissions by 3%–5%; and Simoni and Claudel (2018) developed a hybrid macroscopic traffic model with microscopic delivery vehicle behaviors, finding that 100 delivery vehicles in Austin, TX can change total network delay in the morning hours by 3%. Additionally, two traffic simulation based studies include fuel consumption comparisons, including Chrysostomou et al. (2019), who estimated a 50% increase in fuel consumption due to double parking, and the previously mentioned Alho et al. (2018), who estimated a 68% reduction in fuel consumption with improved loading bay configurations.

Queuing models are also used to estimate congestion effects. Liu et al. (2023) apply a queuing model when the parking demand of ride-hailing vehicles exceeds parking capacity as a method to estimate the spill-over effect of congestion on the surrounding traffic. Delaitre (2009) combines a simulation and queuing model to estimate the number of vehicles obstructed by illegally parked delivery vehicles, reporting that 120 and 312 vehicles were affected from 10 and 11 illegally parked freight vehicles, respectfully. As studied by Portilla et al. (2009), queuing models can also be applied to directly model surrounding traffic vehicle delays due to roadway throughput capacity restrictions from lane obstruction events. Portilla et al. (2009) also presents evidence indicating that queuing models can be a compelling substitute for traffic simulations in certain cases.

To address our research question, we adopt a queuing approach to estimate traffic delay for simplicity and because of the alignment between our optimization model outputs and queuing model inputs. Specifically, we implement a deterministic (D/D/1) queuing model following Mannering and Washburn (2020), described in detail in Section 3.7.

3. Methods

Fig. 2 summarizes our approach to comparing outcomes in coordinated (optimized) reservation systems to status quo uncoordinated (first-come, first-serve) systems, including references to sections and equations that define each process. We begin by defining our approach to modeling uncoordinated and coordinated vehicle arrivals, including the MILP and ILP optimization formulations and their combination into a hybrid approach that avoids the worst case performance in all problem instances tested. We then discuss handling of uncertain arrival times and scheduling buffers. Finally we describe our data, our range of scenarios tested, and our queuing model.

3.1. Uncoordinated vehicle arrivals

Our baseline scenario models uncoordinated delivery vehicle arrivals to a single set of co-located parking spaces, similar to Mor et al. (2020)'s "no booking management" model. For a set of delivery vehicles indexed by i with arrival windows $[a_i, b_i]$ (where we set $a_i = b_i \quad \forall i$ for simplicity) and service duration $s_i > 0 \quad \forall i$, we model uncoordinated arrivals using arrival times t_i, and departure times with $t_i + s_i$. We also assume $t_i = a_i$ in an effort to pick a representative arrival time between a_i and b_i. An illustrative uncoordinated parking schedule is shown in Fig. 3 with four delivery vehicles that arrive at a time when no parking spaces were available and therefore double park for their service duration. Fig. 3 also shows how optimal delivery vehicle scheduling can shift delivery vehicle arrivals based on their parking request windows and reduce double parking.

3.2. Optimally coordinated vehicle arrivals

To model coordinated delivery vehicle arrivals, we develop a hybrid approach leveraging, adapting, and improving both the continuous-time MILP optimization framework developed by Roca-Riu et al. (2015) and the discrete-time ILP formulation developed by Yang et al. (2019) for the parking slot assignment problem. These formulations can generate equivalent solutions when the resolution of the input parameters matches the time discretization resolution (as they do in our problem). We find that the MILP formulation is faster for smaller demand scenarios and the ILP formulation is faster for larger demand scenarios. Our hybrid approach, described in Section 3.3, leverages both formulations to achieve solutions in runtimes that are always faster than the slowest of the two formulations across our runs. We first present the MILP and then the ILP variant.

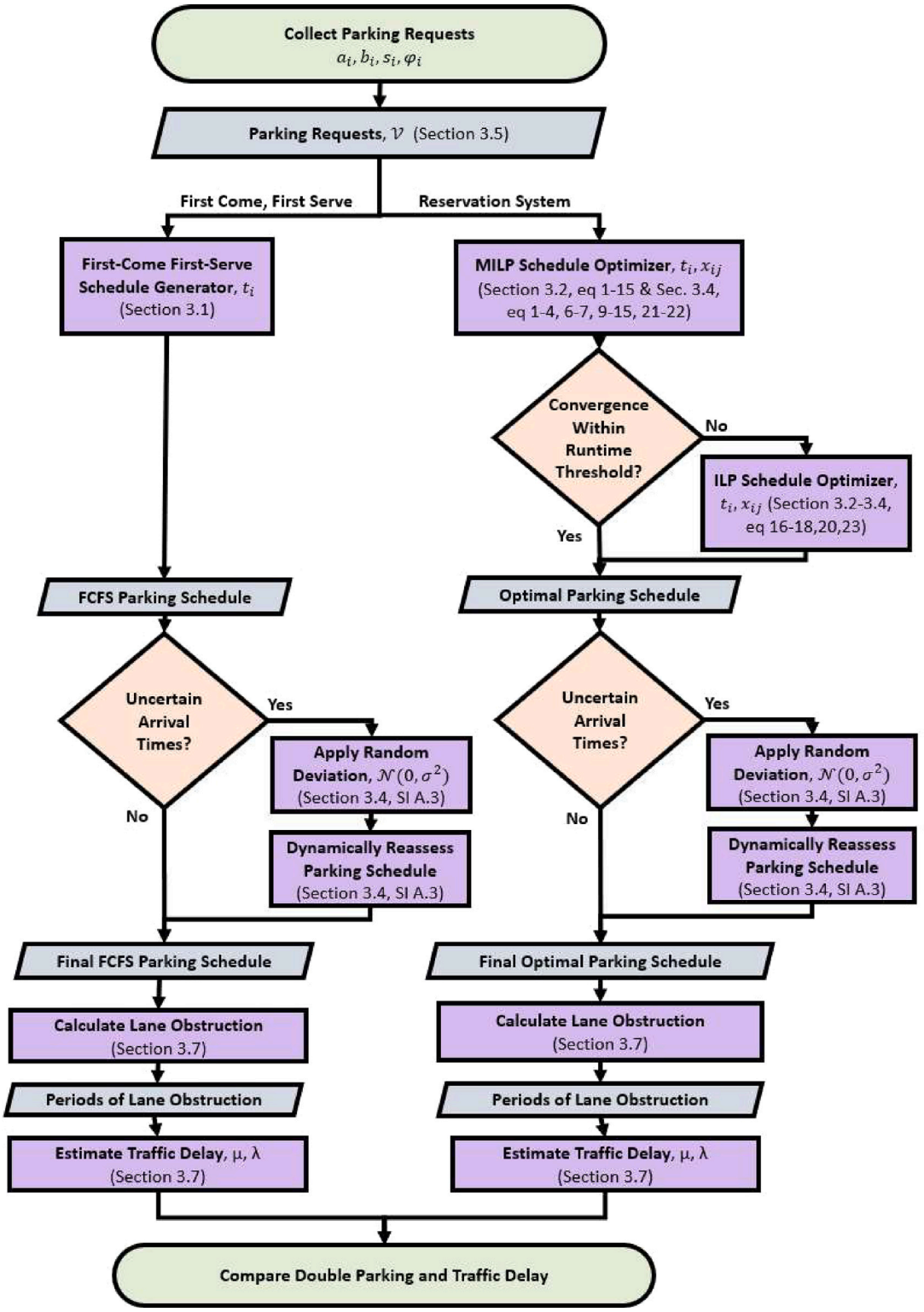


Fig. 2. A high-level summary of our modeling workflow. Key variables, equations and sections are included for reference. Block diagram shapes follow standard conventions, e.g. oval is the start/end, parallelogram represents data, rectangle indicates a process, diamond is a decision node.

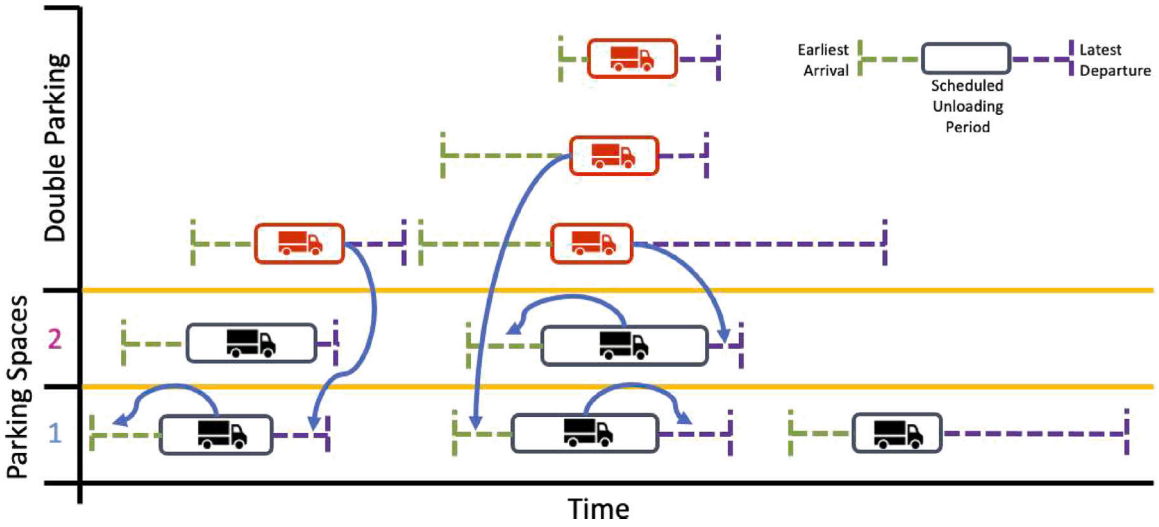


Fig. 3. Illustrative example of uncoordinated and coordinated parking. When uncoordinated, each vehicle arrives at its baseline arrival time. If the available parking spaces are occupied, then the next arriving vehicle will double park for the duration of its service time. Four vehicles double park (colored in red), as shown in the uncoordinated scheduled. In the optimized schedule, arrival times are shifted (blue arrows) within allowable bounds to minimize double parking, resulting in only one vehicle double parked instead of four. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

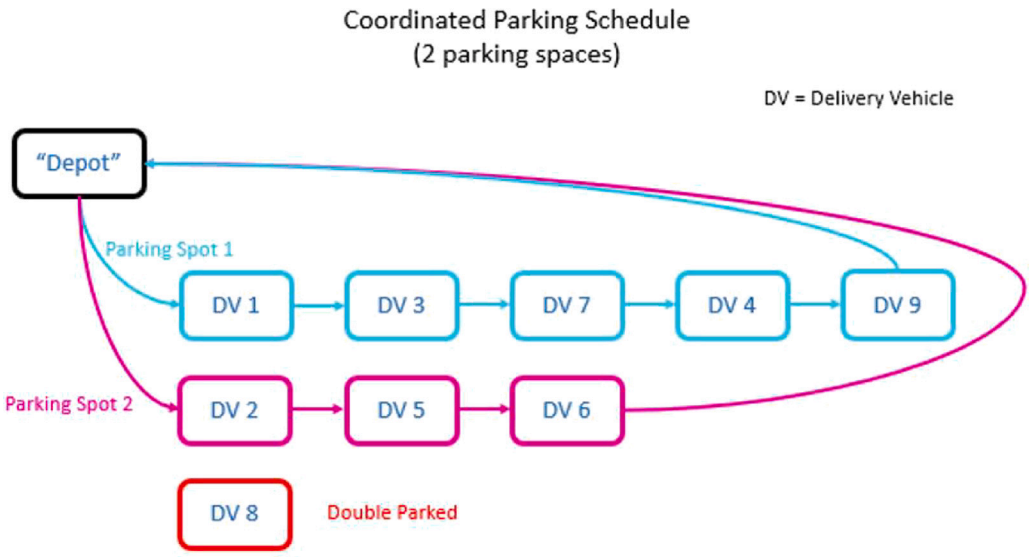


Fig. 4. Illustration of the MILP formulation of the Parking Slot Assignment Problem, adapted from the Vehicle Routing Problem with Time Windows (VRPTW). Each parking space is assigned a sequence of delivery vehicles (DV), analogous to the sequence of locations along a route visited by a vehicle in the VRPTW. The "depot" node in this application serves to limit the number of parallel parking sequences to less than or equal to the number of parking spaces in the scenario, analogous to the starting and ending location for each vehicle in the VRPTW.

MILP formulation

The MILP formulation of the Parking Slot Assignment Problem is based on a formulation of the Vehicle Routing Problem with Time Windows (VRPTW) Desaulniers et al. (2015), where the sequence of vehicles assigned to a specific parking space is analogous to the sequence of delivery locations on a vehicle route in a traditional VRPTW, as illustrated in Fig. 4. The vehicle "depot" node shown in Fig. 4 comes from the VRPTW structure and in this context serves to limit the number of parallel parking sequences to within the parking capacity.

We optimize arrival sequences to minimize the sum of the service durations of unassigned (double-parked) vehicles, adapting Roca-Riu et al. (2015) MOD4b, with modifications as shown below.

$$\text{minimize} \quad \sum_{i \in \mathscr{V}} s_i \beta_i \qquad \text{Minimize unassigned vehicle service duration} \tag{1}$$

subject to

$$\sum_{j \in \mathscr{V}} x_{0j} \leq c \qquad \#\text{ parallel vehicle parking sequences} \leq \text{parking capacity} \tag{2}$$

$$\sum_{(i,j) \in \mathscr{A}} x_{ij} \leq 1, \quad i \in \mathscr{V} \qquad \text{Each vehicle is assigned once or not assigned} \tag{3}$$

$$\sum_{(i,j) \in \mathscr{A}} x_{ij} - \sum_{(j,i) \in \mathscr{A}} x_{ji} = 0, \quad i \in \mathscr{V} \qquad \text{Assigned vehicles are part of a sequence} \tag{4}$$

$$t_j \geq t_i + s_i - (1 - x_{ij}) M_{ij}, \quad i, j \in \mathscr{V} \qquad \text{Prevent overlap in sequential vehicle arrivals} \tag{5}$$

$$t_i \geq a_i - \phi - \beta_i (a_i - \phi - t_{0i}), \quad i \in \mathscr{V} \qquad \text{Each arrival is not before its earliest permitted start time} \tag{6}$$

$$t_i \leq b_i + \phi + \beta_i (t_{0i} - b_i - \phi), \quad i \in \mathscr{V} \qquad \text{Each arrival is not after its latest permitted start time} \tag{7}$$

$$t_i + s_i \leq t_{\text{END}}, \quad i \in \mathscr{V} \qquad \text{Vehicle must complete service before end of scenario} \tag{8}$$

$$t_i \geq t_{\text{START}}, \quad i \in \mathscr{V} \qquad \text{Vehicle must arrive after the start of the scenario} \tag{9}$$

$$\sum_{i \in \mathscr{V}} s_i \beta_i \geq 0 \qquad \text{Valid cut to improve efficiency} \tag{10}$$

$$t_i \in \mathbb{R}, \quad i \in \mathscr{V} \qquad \text{Scheduled arrival time for each vehicle} \tag{11}$$

$$x_{ij} \in \{0, 1\}, \quad (i, j) \in \mathscr{A} \qquad 1 \text{ iff vehicle } j \text{ follows } i \text{ in parking sequence} \tag{12}$$

where

$$\beta_i = \left(1 - \sum_{(i,j) \in \mathscr{A}} x_{ij}\right), \quad i \in \mathscr{V} \qquad 0 \text{ if vehicle } i \text{ is assigned a space, 1 if it double parks} \tag{13}$$

$$t_{0i} = a_i, \quad i \in \mathscr{V} \qquad \text{Arrival time for unscheduled vehicles (for efficiency)} \tag{14}$$

$$M_{ij} = \min(t_{\text{END}}, \ b_i + s_i + \phi) - \max(t_{\text{START}}, \ a_j - \phi) \qquad \text{Smallest "Big M" value that does not constrain } t_i \text{ or } t_j \tag{15}$$

where $i \in \mathscr{V}$ indexes the set of delivery vehicles (as well as a "depot" node $i = 0$ that serves as the start and end node of each parking sequence); $(i, j) \in \mathscr{A}$ is the set of arcs in a directed network; x_{ij} is a binary decision variable that is equal to 1 if vehicle i is scheduled immediately before vehicle j in the same parking spot (0 otherwise); t_i is a continuous decision variable that represents the scheduled start time of delivery vehicle i; t_{0i} is the arrival time value used for vehicles that are not assigned (imposed to improve efficiency); t_{START} and t_{END} are the start and end of the full scenario window; c is the parking capacity (number of parking spaces); a_i and b_i are the bounds of the requested arrival time window for vehicle i; s_i is the service duration for vehicle i; ϕ is the maximum permitted shift in arrival time beyond the requested arrival time window for all vehicles (which we use for parametric study); and M_{ij} is a constant for the "Big M" representation of a disjunctive constraint, using the tightest value that does not remove any of the feasible domain. More specifically, the value of M_{ij} is the smallest value that avoids constraining t_i and t_j when $x_{ij} = 0$: $M_{ij} = \min(t_{\text{END}}, \ b_i + s_i + \phi) - \max(t_{\text{START}}, \ a_j + \phi)$. Additionally, in our simulations we remove and redraw any delivery vehicle parking requests where the sum of the requested arrival time and service time, $a_i + s_i > t_{\text{END}}$. This prevents providing an infeasible delivery vehicle parking request for both uncoordinated and coordinated schedules. For efficiency and stability, we scale constraints with t_i by the length of the scenario (divide both sides of Eqs. (5)–(9) by $t_{\text{END}} - t_{\text{START}}$) and we scale the objective function by the average of the service durations, $\sum_i s_i / |\mathscr{V}|$.

Relative to the formulation in Roca-Riu et al. (2015), our formulation eliminates a vector of decision variables, modifies constraints to align with our scenario of interest and improve efficiency, and changes the domain of several variables. First, we remove the binary decision β_i and replaced it with a definition that is a function of the vehicle scheduling decision variables x_{ij} to reduce problem dimensionality and improve efficiency (see definition Eq. (13)). We also relax constraint Eq. (3) so that we no longer require every delivery vehicle to be assigned a parking space, ideally reducing algorithm complexity. Next, we modify the time window constraints Eqs. (6) and (7) to conform with unassigned vehicles, remove a "Big M" constraint, and allow for arrival request flexibility, ϕ. Furthermore, the time window constraints also default to t_{0i} if the delivery vehicle is unassigned, as defined in Eq. (14) and used in constraints Eqs. (6)–(7), in order to restrict the search space and improve efficiency. We also include the service time in constraint Eq. (8) to prevent vehicles from being scheduled such that they might be parked beyond the end of the scenario time window. We also redefine the remaining "Big M" variable from a common value across all delivery vehicles to the smallest value specific for each delivery vehicle sequence pair that does not remove any of the feasible region, in Eq. (15). Additionally, we add a valid cut to improve efficiency by bounding the objective function in constraint Eq. (10). Finally, in our simulations, we ensure that inputs a_i, b_i, s_i and ϕ are integers in order to align with the inputs of the ILP variant.

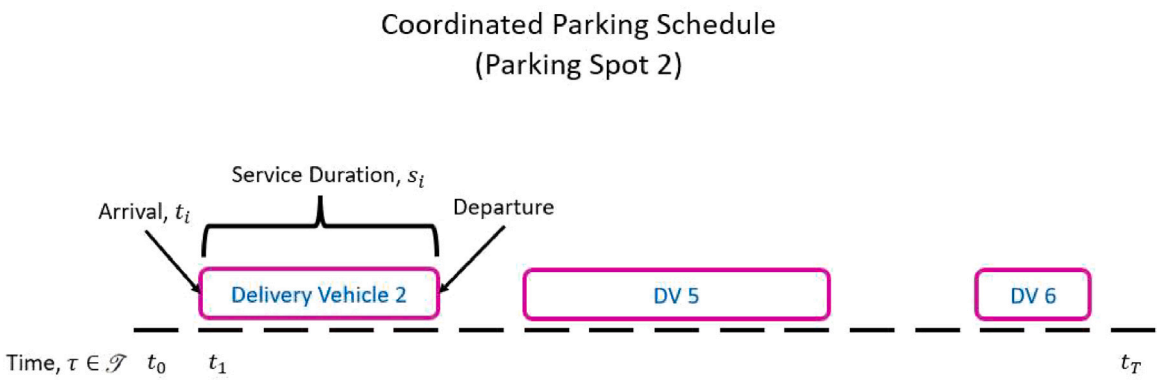


Fig. 5. Illustration of the discrete-time ILP Parking Slot Assignment Problem for Parking Spot 2 from Fig. 4. Time is discretized into τ steps within $\mathcal{T}$. Each delivery vehicle is assigned an arrival time such that overlap between the vehicles does not exceeds the number of parking spaces at any time step.

ILP formulation

In the ILP formulation of the Parking Slot Assignment Problem time is discretized into time steps $\tau \in \mathcal{T}$. Each time step is available for scheduling vehicle arrivals, $i \in \mathcal{V}$. Fig. 5 illustrates a solution from the ILP model for Parking Spot 2 from the previous example in Fig. 4.

In the ILP formulation, we represent delivery vehicle parking requests with a service start time and duration vector that corresponds to the bid vector in Yang et al. (2019). The full scenario scheduling window $t \in [t_{\text{START}}, t_{\text{END}}] \subseteq \mathbb{R}$ is divided into discrete time steps indexed by $\tau \in \mathcal{T} = \{0, 1, 2, \dots, T\}$, each of which has an associated time $t_\tau \in \mathbb{Z}$, where $t_0 = t_{\text{START}}$ and $t_T = t_{\text{END}}$. Each vehicle i from the set of delivery vehicles $\mathcal{V}$ has a requested service duration $s_i \in \mathbb{Z}$, requested arrival time $a_i \in \mathbb{Z}$, and arrival time flexibility $\phi \in \mathbb{Z}$. These requests are encoded in a discretized service request vector $\mathbf{s}_i = [s_{i0}, s_{i1}, \dots, s_{iT}]^\top$, whose elements $s_{i\tau}$ have values $s_{i\tau} = s_i$ for all $\{\tau : a_i - \phi \le t_\tau \le a_i + \phi\}$ and $s_{i\tau} = 0$ for all other time steps τ. The ILP formulation is:

$$\text{maximize} \sum_{i \in \mathcal{V}} \sum_{\tau \in \mathcal{T}} s_{i\tau} x_{i\tau} \qquad \text{Maximize fulfilled vehicle service duration} \tag{16}$$

$$\text{subject to} \sum_{\tau \in \mathcal{T}} x_{i\tau} \le 1, \; i \in \mathcal{V} \qquad \text{A delivery vehicle can be assigned once or not at all} \tag{17}$$

$$\sum_{\tau \in \mathcal{W}_i} x_{i\tau} = 0, \; i \in \mathcal{V} \qquad \text{Prevent arrivals where service exceeds end of scenario} \tag{18}$$

$$\sum_{i \in \mathcal{V}} \sum_{k \in \mathcal{T}_{i\tau}} x_{ik} \le c, \quad \tau \in \mathcal{T} \qquad \text{Limit occupancy given prior arrivals and service durations} \tag{19}$$

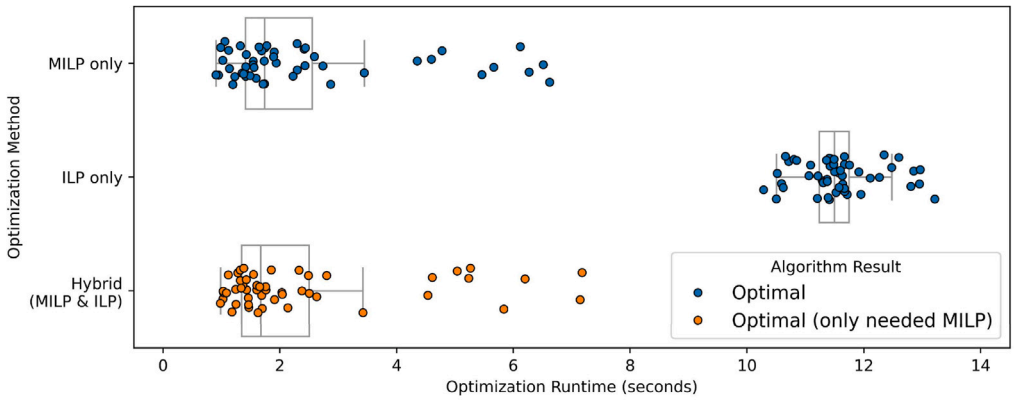
$$x_{i\tau} \in \{0, 1\}, \; i \in \mathcal{V}, \; \tau \in \mathcal{T} \qquad 1 \text{ iff delivery vehicle } i \text{ is assigned to arrive at time step } \tau \tag{20}$$

where $i \in \mathcal{V}$ indexes the set of delivery vehicles, $\tau \in \mathcal{T}$ indexes the discretized time steps, $x_{i\tau}$ is a binary decision variable that is equal to 1 if delivery vehicle i is assigned to arrive at time step τ (0 otherwise), c is the number of parking spaces, $\mathcal{W}_i = \{\tau : \tau \in \mathcal{T}, \; t_\tau > t_{END} - s_i\}$ is the set of arrival time steps for which vehicle i's service duration would exceed the end of the full scenario scheduling window, and $\mathcal{T}_{i\tau} = \{k : k \in \mathcal{T}, \; t_\tau - s_i < t_k \le t_\tau\}$ is the set of time steps within vehicle i's service duration window preceding time step τ.[2] We modify the Yang formulation by adding Eq. (18) to prevent scheduled arrivals for service durations that exceed the end of the scenario window.

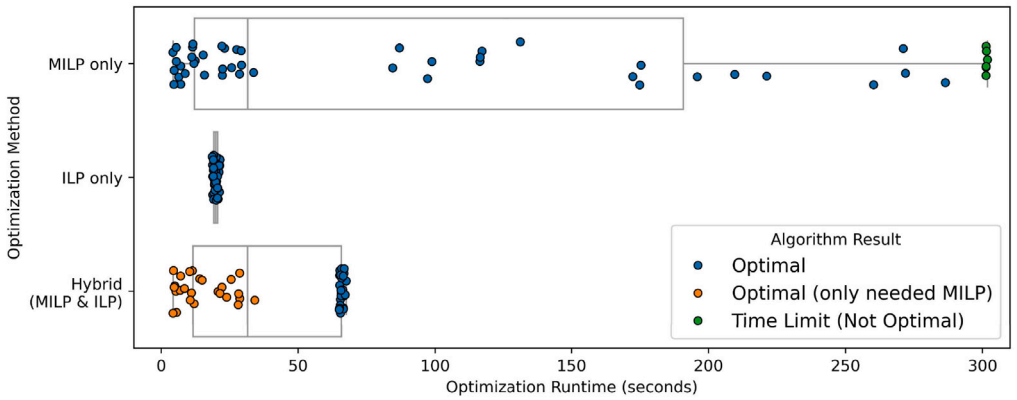
3.3. Hybrid MILP/ILP approach

A summary of runtime results across three scenarios is shown in Fig. 6. We find that our MILP formulation typically solves problem instances with lower demand faster than our ILP formulation because of the time that it takes the ILP to load the problem into memory. However, the MILP method has a more variable computational time, and for some problem instances it can require dramatically higher computational time, potentially failing to reach a solution in a reasonable time. The ILP formulation typically solves problem instances with higher demand more efficiently; however, scalability may be limited by available memory. We leverage both formulations by solving each problem instance first with the MILP formulation but switching to the ILP formulation

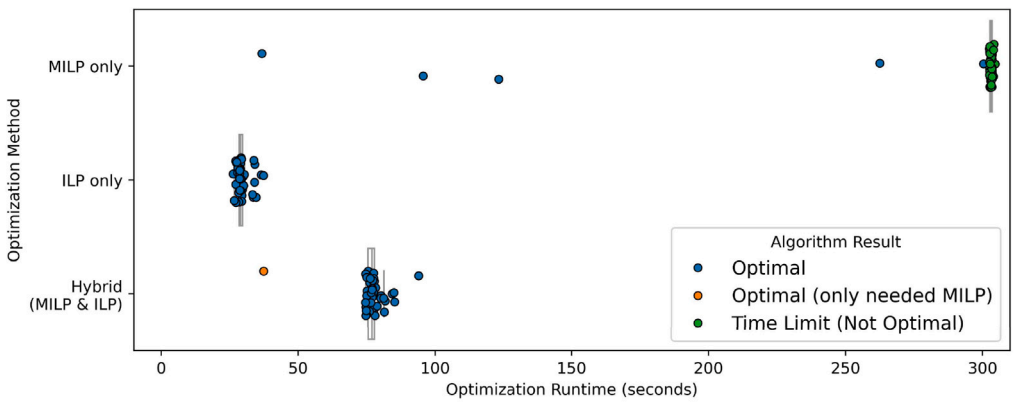
[2] In the original Ref. Yang et al. (2019), $\mathcal{T}_{i\tau}$ was defined to exclude time step τ. After correspondence with the authors we determined this to be a typo, and we include time step τ.



(a) Low Demand (3 spaces, 2 vehicles/hr/space)



(b) Medium Demand (3 spaces, 3 vehicles/hr/space)



(c) High Demand (3 spaces, 4 vehicles/hr/space)

Fig. 6. Algorithm runtime comparison in low, medium, and high demand scenarios, including time to load the problem into memory and execute the solver. All cases use a 5 min buffer between each reservation (ψ) and ±5 min of permitted shift in arrival time (ϕ). The maximum runtime on the "MILP only" method was limited to 300 s for this demonstration. Individual dots represent individual problem instances (draws from the delivery timing request data).

if the MILP execution time exceeds a user-specified threshold (45 s in our runs).[3] This hybrid approach outperforms either the MILP or the ILP formulation in every instance tested, avoiding worst-case performance for every problem instance.

3.4. Arrival time uncertainty

Following the prior literature, we assume in our base optimization formulations that delivery vehicles will arrive exactly as they are scheduled, representing an idealized case; however, in a real-world application there is typically uncertainty of vehicle arrival time. Jiang and Fan (2020) explored the effects of uncertain arrival and departure time with their optimization formulation that includes a "parking unpunctuality time window" and they found that planning ahead for stochastic vehicle behaviors reduces parking space utilization due to needing more idle parking spaces to support demand. To test the robustness of our results to uncertain arrival times, we add a parameter ψ to the optimization structures to represent a minimum required buffer between the end of one reservation window and the beginning of the next reservation window scheduled for the same parking space. A description of how the buffer is incorporated in both formulations is discussed below. Furthermore, our results include outputs from both the idealized and more realistic model with uncertainty in arrival times and buffers between reservations.

Optimization with schedule buffer

The new buffer variable ψ is intended to leave extra time in the schedule between one vehicle request to the next so that random variation in arrivals can be accommodated while guaranteeing that scheduled vehicles who arrive up to ψ min after their scheduled arrival time will have their reserved space available.

For the MILP formulation, this can be accomplished by adapting Eqs. (5) to (21) and (8) to (22). The definition of M_{ij} will also include ψ such that $M_{ij} = \min(t_{\text{END}},\ b_i + \phi + s_i + \psi) - \max(t_{\text{START}},\ a_j + \phi)$.

$$t_j \geq t_i + s_i + \psi - (1 - x_{ij})M_{ij},\quad i, j \in \mathscr{V} \qquad \text{Prevent overlap in sequential arrivals} \tag{21}$$

$$t_i + s_i + \psi \leq t_{\text{END}},\quad i \in \mathscr{V} \qquad \text{Must complete service before end of scenario} \tag{22}$$

The ILP formulation also requires changing $\mathscr{T}_{i\tau}$ in Eq. (19) to definition (23).

$$\mathscr{T}_{i\tau} = \{k : k \in \mathscr{T},\ t_\tau - s_i - \psi < t_k \leq t_\tau\} \qquad \text{Limit occupancy given prior arrivals and service} \tag{23}$$

Adding random deviation to vehicle arrivals

The second step in our robustness check is to randomly shift delivery vehicle arrivals. We apply cases with a normal distribution with a mean of zero and a standard deviation of σ, resulting in 95% of vehicles arriving within roughly a 4σ-min window surrounding the scheduled time. We test values of $\sigma = \{2, 7\}$, corresponding to 95% arrival windows of 8 and 28 min, respectively, surrounding the target arrival time. After shifting the arrivals, we dynamically reassess which vehicles are able to park legally and which will be forced to double parked for both uncoordinated and coordinated schedules. A more detailed description of the reassessment workflow of the coordinated schedule is included in Appendix Section C.

3.5. Dataset

Our dataset requires delivery vehicle specific arrival request windows, from time a to b, and service or parking duration s. However, requested delivery windows cannot be captured through empirical observation alone and would require in-depth delivery company scheduling information which was not available for this research. Instead, we assume that $a = b = t$ (where t is the empirically observed arrival time of a delivery vehicle) and the span of the requested time window can be expanded with parameter ϕ, allowable schedule flexibility. The set of delivery vehicle arrival requests is enumerated in the set $\mathscr{V}$. In an uncoordinated scenario, we use these data to derive vehicle arrival time, t and departure, $t + s$. In the coordinated scenario, a, ϕ and s are inputs to Eqs. (1)–(15) and/or Eqs. (16)–(20), each of which return an optimal schedule of delivery vehicle arrivals.

We examine two datasets in this research. The first dataset is derived from empirical data collected by commercial company Coord over the first 100 days of their Smart Zone pilot program in Aspen, CO, from November 2020 through January 2021 Coord (2021b). The pilot program implemented a loading zone reservation system for a set of curbside and alley commercial loading zones and supported 1520 parking sessions. Using the Aspen dataset, which provides data at an hourly resolution, we determined the set of requested vehicle arrival times a by sampling the empirical frequency distribution for vehicle arrivals by hour and adding a random value uniformly drawn from between 0 and 60 min. For service duration s, we sampled from a normal distribution with a heterogeneous mean parameter dependent on the hour of the vehicle arrival and with a 35 min mean parking duration across the entire as provided by Coord (2021b). We are, however, missing data on the variance associated with each of the average service duration data points, so we follow Roca-Riu et al. (2015) in assuming a standard deviation of five minutes for each hour (based on empirical observations of Barcelona, Spain in 1997). Of note, a_i, s_i are each rounded to the nearest integer to align inputs for both versions of the parking slot assignment problem.

[3] The 45 s threshold is variable based on the capabilities of the computer where a better equipped machine might extend the performance of MILP or reduce memory storage time in the ILP. To improve our runtime in the future we may even consider a smaller threshold based on the short memory storage time of the ILP shown in Fig. 6.

Our second dataset was collected by Street Sense Inc. and their camera monitoring and machine learning characterization system on Liberty Ave in downtown Pittsburgh from September to November of 2021. Over this period, Street Sense Inc. characterized more than 6500 cars and 500 trucks both parked legally in parking spaces and illegally double parked in a lane of traffic. For our study we focus on a subset of truck parking observations from which we randomly sampled to create a set of vehicle arrival schedules. The subset of data is described in more detail in Appendix Section B. One limitation is that the Street Sense algorithm broadly characterizes trucks to include not only delivery vehicles, but also passenger pickup trucks. The Aspen dataset contains data from 7 am–6 pm, an 11 h time window, and we select a 20 h time window from the Pittsburgh dataset which captures the majority of the delivery vehicle arrival activity from 4 am — midnight.

3.6. Delivery vehicle scheduling scenarios

Our research explored combinations of three independent variables including: the number of co-located parking spaces $c \in \{1, 2, \ldots, 7\}$, the number of delivery vehicles n, as described in more detail below, and the minutes of allowable arrival flexibility to parking requests $\phi \in \{0, 1, 2, 5, 10, 15, 30\}$ min. We determined the number of delivery vehicles n by first considering the expected hourly demand per parking space ranging from a single delivery vehicle per day to a maximum of 6 delivery vehicles per parking space per hour. For example, a smart curbspace system with 7 parking spaces and a demand of 6 delivery vehicles per parking space per hour across the Aspen 11 h scenario window results in a maximum of 462 delivery vehicles per day, whereas the maximum number of delivery vehicles for a single parking space smart curb system is 66. To explore a range of scenarios, we uniformly sampled 100 different values for n between 1 and the upper bound of the number of delivery vehicles for each of the 1–7 parking space configurations.

Both the MILP formulation and the ILP formulation experience long processing times under certain scenarios. To leverage the advantages of both formulations and improve computational tractability, we employed several strategies. First, to decrease MILP processing time, we initialized the solver with a solution from the previous parking slot assignment problem iteration when possible, e.g. in our algorithm we establish the vehicle requests and then execute the parking slot assignment problem over sequential steps in ϕ. We also set the relative tolerance gap on the bounds of the solution to 1% for the MILP and ILP, which had a notable reduction on processing time relative to the default 0.01% tolerance but allowed for slightly sub-optimal solutions.

3.7. Double parking and deterministic queuing

In order to assess the potential benefits of a reservable curbspace system, we implemented a D/D/1 deterministic arrival and departure, single server queuing model, as described in Mannering and Washburn (2020). The primary inputs to our queuing model are the lane obstruction events experienced during periods of double parked vehicles at the reservable curbspace location. These double parked vehicles are a result of vehicle requests that remain unassigned and not given a parking space in the optimal parking schedule. We use the periods of lane obstruction to model reduced traffic flow near the reservable curbspace location that can cause vehicles delay and unnecessary fuel consumption.

Our queuing model consists of a two-lane, same direction flow of traffic, downtown city street past the reservable curbspace location where surrounding vehicle traffic arrives and departs with uniform rates. In the model, we assume that double parked vehicles will only disrupt a single lane of traffic flow regardless of the number of double parked vehicles. Additionally, we limit the maximum queue length to 30 vehicles based on the number of cars in a city block.[4] Implementing a maximum queue length aims to capture the effects of vehicles that re-route around a long queue, rather than waiting in the queue, without explicitly modeling routing or rerouting behavior.

Fig. 7 provides an illustration of a queuing model with arrival and departure curves over time. In certain scenarios, where during a period of lane obstruction the departure rate is less than the arrival rate, a queue can form that is reflected by the difference between the two curves. The area of this difference over time (shown in gray) is equivalent to total traffic delay. If we assume that the vehicles in the queue are idling, then total fuel consumption can be estimated by scaling the total minutes of vehicle delay by $\frac{1}{240}$ gallons per minute.[5] Additional information on the queuing model is discussed in Appendix Section D including our complete D/D/1 queuing workflow in Figure D.3.

Our presented results are based on downtown traffic flow information from Aspen, CO and Pittsburgh, PA as described in Colorado Department of Transportation (2020), PennDOT (2021) and street capacity estimates from the Highway Capacity Manual 2000 Transportation Research Board Committee on Highway Capacity and Quality of Service (2000). We examine a base case along with optimistic and pessimistic traffic flow scenarios as shown in Table 2 with additional details on the basis for these inputs provided in Appendix Section A.

4. Results

The following section contains a comparison between uncoordinated (first-come, first-serve) and coordinated (optimal) vehicle arrivals in terms of double parking events and the associated impact on fuel consumption. We conducted our analysis with Python v3.10.4 and the Gurobi solver v9.5.1.

[4] ~22 ft/car NADA Guides (2021) and 650 ft/Chicago city block TheColdWire.com (2021).

[5] $\sim\frac{1}{4}$ of a gallon of gas is lost for an hour of idling, a middle estimate between compact and large sedans Energy.gov (2015).

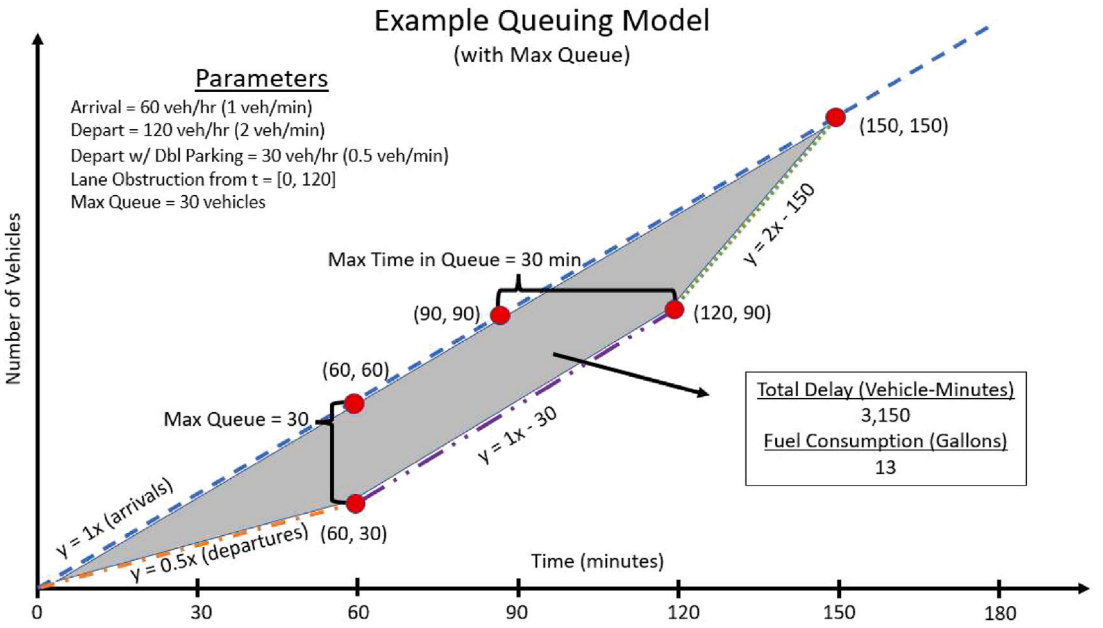


Fig. 7. An illustrative example deterministic queuing model with lane obstruction from time 0 until time 120 min and a maximum queue of 30 vehicles. In this example, the vehicle departure rate is less than the arrival rate due to the period of lane obstruction. This creates an area between the two curves that can be interpreted as our primary output metric of the total vehicle-minutes of delay.

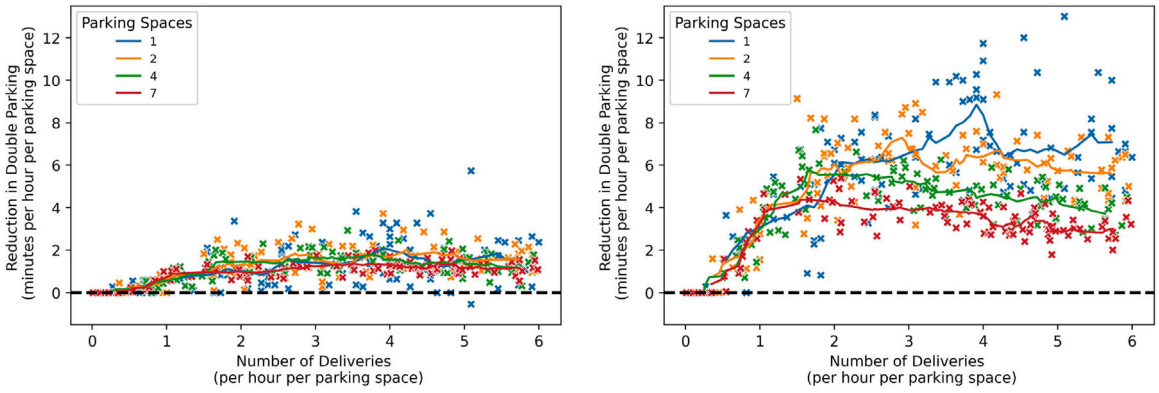
Table 2
Summary of vehicle arrival and departure rates used as inputs to the queuing model (vehicles/hr).

Scenario	Arrival rate	Departure rate (No obstruction)	Departure rate (Lane obstructed)
Optimistic	375	3820	1528
Pessimistic	1833	2136	534
Base Case	1104	2582	861

4.1. Double parking reduction

Our first comparison is the reduction in double parking time at a set of co-located parking spaces per hour per parking space from uncoordinated to coordinated arrival schedules given zero and five minutes of arrival flexibility and no uncertainty in arrival times. Results from our Aspen dataset are shown in Fig. 8. The results show that reductions vary substantially across cases, but, on average, optimal scheduling can reduce total double parking by about 1 to 2 min per parking space per hour (for arrival rates greater than 1 vehicle per hour) even when vehicles have no arrival time flexibility ($\phi = 0$). This is due to solver flexibility to choose which vehicles double park rather than being restricted to first-come, first-serve. Including ±5 min of arrival time flexibility ($\phi = 5$) can reduce average double parking by 3 to 8 min per hour per parking space. Furthermore, the reduction in double parking appears to plateau and then slowly drop with increasing quantities of delivery vehicles as parking resources become saturated and parking capacity utilization approaches 100% in both uncoordinated and coordinated cases. The reduction in double parking for other values of $\phi \in \{0, 1, 2, 5, 10, 15, 30\}$ are available in Appendix Figure G.1 and Appendix Figure H.1 for the Aspen and Pittsburgh datasets, respectfully.

We also examined a measure of effectiveness of a curbspace reservation system by comparing the reduction in double parking time from curbspace reservation conversion to the reduction from parking capacity expansion. The Pittsburgh dataset resulted in larger ratios that favor a curbspace reservation system, and the graphic is shown in Fig. 9 (the Aspen ratios are shown in Appendix Figure I.1). Here, each boxplot represents a distribution of the reduction in double parking ratios between the options across our delivery vehicle scenarios. The red dashed line indicates equal reduction from adding one first-come, first-serve parking space and from converting existing spaces to a reservation system, and any ratio above the line indicates that reservation system conversion is reducing double parking time more than parking space expansion. In general, we see that converting curbspace to a reservation system can provide the equivalent reduction in double parking as expanded parking capacity in some cases. Additionally, we see that curbspace reservation effectiveness increases relative to adding another parking space as the arrival flexibility increases, as expected.



(a) $\phi = 0$ minutes of scheduled arrival time flexibility (b) $\phi = 5$ minutes of scheduled arrival time flexibility

Fig. 8. A comparison of the reduction in double parking duration for a set of co-located parking spaces per hour per parking space with $\phi = 0$ and $\phi = 5$ min of arrival flexibility, with no arrival uncertainty and $\psi = 0$ min of buffer between reservations. The displayed trend lines are based on an 11-sample rolling average. Of note, there is one instance of negative reduction in double parking with the optimized parking schedule (data below the dashed line) when $\phi = 0$. This occurs due to the use of a 1% optimality tolerance, that allows the algorithm to terminate at a slightly sub-optimal solution. We accept this level of tolerance to improve runtime.

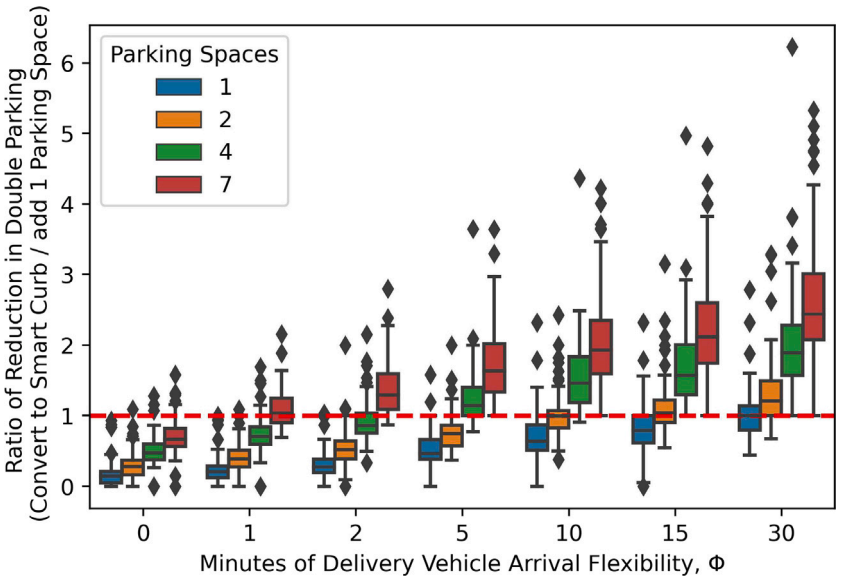


Fig. 9. Parking capacity expansion equivalence of double-parking reductions via reservable curb management. Box plots show the ratio, across a range of delivery vehicle scenarios, of (1) the reduction in double parking generated by converting from uncoordinated parking spaces to reservable curb spaces to (2) the reduction in double parking generated by adding one additional uncoordinated space. Any ratio above the red dashed line indicates the reservable curbspace system provides more double parking reduction than adding an extra parking space. Ratios are based on the Pittsburgh dataset assuming no uncertainty in arrival time, requiring no buffers between reservations and with co-located parking spaces. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.2. Uncertain arrival times and schedule buffers

As mentioned in Section 3.4, the results from our optimization model presented so far represent an ideal case where we assume that vehicles will arrive precisely when they are scheduled (e.g. no arrival uncertainty), allowing for tightly scheduled reservations. To add realism, we introduce a buffer parameter ψ to enforce a minimum time between reservations in the optimal schedule. To test the robustness of the optimal schedule with a buffer between reservations, we randomly shift delivery vehicle arrivals by amounts randomly drawn from $\mathcal{N}(0, \sigma^2)$ after the optimal schedule is generated, as described in Section 3.4, and then reassess the sequence of delivery vehicle arrivals to identify which vehicles are accommodated and which must double park. Adding an appropriate buffer duration to the schedule intends to allow originally planned legal parked delivery vehicles to retain their parking space reservation even if they arrive slightly early or late, but vehicles that arrive later than the buffer allows risk losing their space, and vehicles

that arrive earlier than the reservation time may wait for the reservation to begin or double park if the wait time is too long. An additional description of the schedule reassessment process is available in Appendix Section C. Fig. 10 displays the reduction in double parking metric in an ideal case and a case with buffers and arrival time uncertainty for both the Aspen and Pittsburgh datasets.

Specifically, Fig. 10 compares cases with and without a $\psi = 5$ min buffer using both the Pittsburgh and Aspen data when actual arrival times are randomly shifted by an amount drawn from a normal distribution $\mathcal{N}(0, \sigma^2)$ with mean zero and standard deviation $\sigma = 2$. Fig. 10(a) is the reduction in double parking without a buffer from the ideal case for comparison, (identical to Fig. 8(b), but re-scaled). Fig. 10(b) shows how adding a five minute buffer and $\mathcal{N}(0, 4)$ random error in arrival time reduces the effectiveness of a reservable curbspace system relative to uncoordinated first-come, first-serve delivery vehicle arrivals. This is due to the inherent loss of available scheduling time that has been allocated to the buffer between reservations in the optimal schedule that is not imposed on the uncoordinated delivery vehicle arrival schedule as well as inefficiencies in optimal scheduling when arrival times are uncertain. A similar conclusion is reached by Jiang and Fan (2020), where they find that parking space scheduling with arrival and departure uncertainty increases the amount of idle parking space held in reserve that therefore reduces the ability to serve more customers. Furthermore, in some of our test cases we observe negative reductions in double parking, which indicates that the uncoordinated arrival schedule outperforms the optimally scheduled solution subject to buffer constraints and arrival time uncertainty. Figs. 10(c) and 10(d) present the same comparison for the Pittsburgh dataset. Different from the Aspen data, we do not see negative reductions in double parking. This may be due to the generally shorter service duration of the vehicles in the Pittsburgh dataset compared to the Aspen dataset, which allows for more resiliency and the incorporation of new vehicles during the random shift and reassessment process. For example, these short duration curbspace users may be able to take advantage of short gaps created by the schedule buffer and randomly shifted arrivals more often than longer duration delivery vehicle parking requests.

In summary, adding buffers and uncertain arrival times reduces the effectiveness of a reservable curbspace system. This has a dramatic effect for vehicle delivery demand in our Aspen dataset and changes our initial reduction estimate from 3 to 8 min per hour per parking space to −2 to 2 min per hour per parking space. Similarly, estimates for the Pittsburgh dataset are reduced from 5 to 7 min per hour per parking space down to 2 to 5 min per hour per parking space. An additional investigation with 15 min buffers for Aspen and Pittsburgh is available in Appendix Section F and shows an even larger reduction in effectiveness.

4.3. Effects on congestion and energy consumption

Next, we examine congestion and energy consumption. As described previously, we base our queuing model for estimating congestion and energy consumption effects on lane obstruction events (time during which at least one vehicle is double parked), noting that the optimal delivery vehicle schedule that reduces total double parking vehicle time may not be the optimal solution for reducing lane obstruction time.

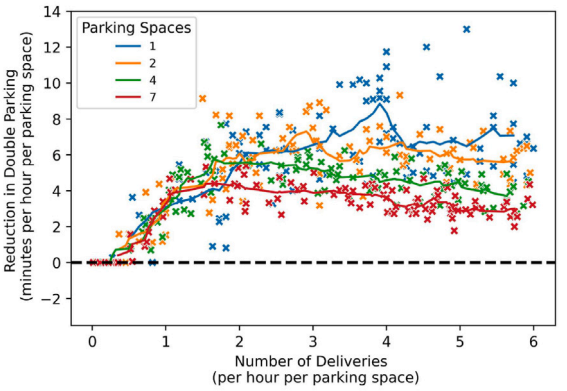
Fig. 11 displays the highest and lowest observed annual reduction in total vehicle hours of vehicle delay and gallons of fuel lost per year from uncoordinated to coordinated arrivals to a set of co-located parking spaces across cases with ±5 min of arrival flexibility, 5 or 15 min buffers between reservations, actual arrival times that differ from scheduled arrival times by amounts randomly drawn from $\mathcal{N}(0, \sigma^2)$ with $\sigma = 2$ or 7 and base case traffic scenarios. The annual estimates assume 365 days of reservable curbspace implementation and similar delivery vehicle demand (vehicle demand arrival patterns were averaged across weekdays and weekends in the data). A complete set of graphics based on reductions per hour across traffic scenarios (Table 2) is shown in Appendix Section J. Additionally, a set of graphics normalized by hour and number of parking spaces across traffic scenarios is provided in Appendix Section K. In both Appendix sections, the optimistic scenarios do not result in any lane obstruction and therefore there is no reduction between first-come, first-serve and optimal cases. Furthermore, the pessimistic traffic scenario results are similar to the base case traffic scenario, which is described in more detail below.

Reductions vary significantly across scenarios (marked as ×'s), and we see generally zero or negative reductions in the worst case and positive reductions in the best case. In the worst case, a single optimized parking space can result in a rolling average increase (negative reduction) of almost 4000 h of vehicle delay per year and about 1000 gallons of fuel relative to an uncoordinated space. In the best case a single parking space reaches a rolling average reduction of 12,000 h of vehicle delay per year and approximately 3000 gallon of fuel consumed per year with diminishing returns as more parking spaces are involved.

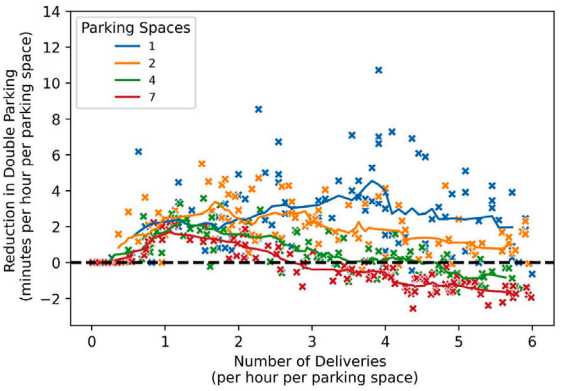
5. Discussion

5.1. Interpretation of results

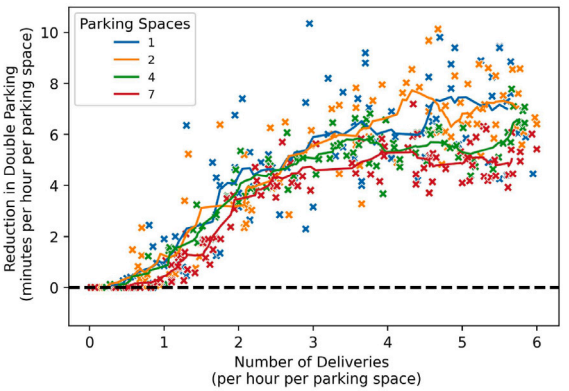
Curbspace management is a pressing issue for cities of all sizes. Demand for curb space is increasing, with annual US parcel delivery volume doubling from 10 billion to 20 billion packages from 2015 to 2020 Spadafora and Rodriguez (2021), and traffic congestion caused an estimated 4.3 billion hours of traffic delay and 1.7 billion gallons of wasted fuel in 2020, despite the pandemic Schrank et al. (2021). Additionally, Jaller et al. (2023) model the potential projection of shopping trends by 2050 across major six major metropolitan areas in the US finding that while CO2 emissions and vehicle miles travel vary based on city population, demographic, and common travel mode choice, new transportation technologies and polices are important for controlling negative externalities. In this research, we characterize the potential for curb reservation systems to reduce negative externalities such as double parking and associated energy consumption, emissions and traffic congestion under ideal and realistic conditions.



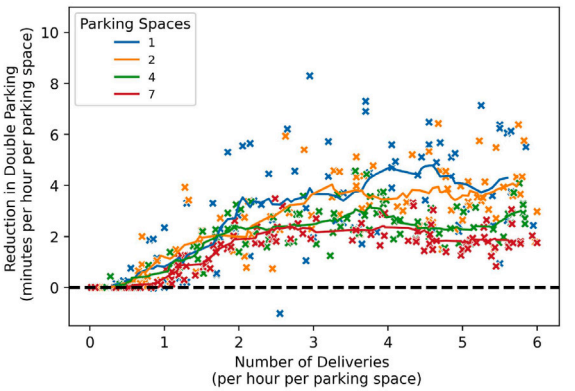
(a) **Aspen ideal case**
$\phi = 5$ minutes of scheduled arrival time flexibility
$\psi = 0$ minutes of buffer required between reservations
$\sigma = 0$ standard deviation of uncertain arrival time



(b) **Aspen with uncertain arrival times**
$\phi = 5$ minutes of scheduled arrival time flexibility
$\psi = 5$ minutes of buffer required between reservations
$\sigma = 2$ standard deviation of uncertain arrival time



(c) **Pittsburgh ideal case**
$\phi = 5$ minutes of scheduled arrival time flexibility
$\psi = 0$ minutes of buffer required between reservations
$\sigma = 0$ standard deviation of uncertain arrival time

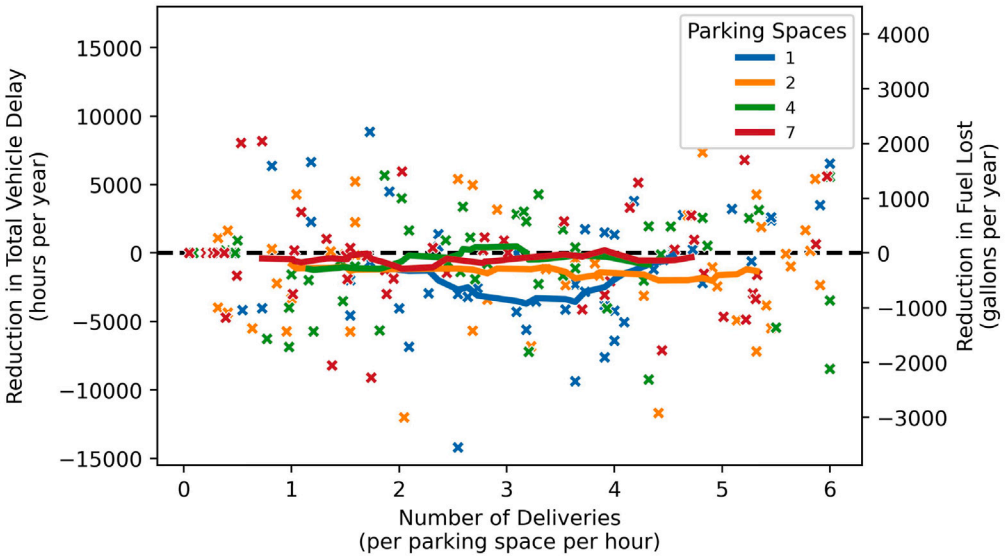


(d) **Pittsburgh with uncertain arrival times**
$\phi = 5$ minutes of scheduled arrival time flexibility
$\psi = 5$ minutes of buffer required between reservations
$\sigma = 2$ standard deviation of uncertain arrival time

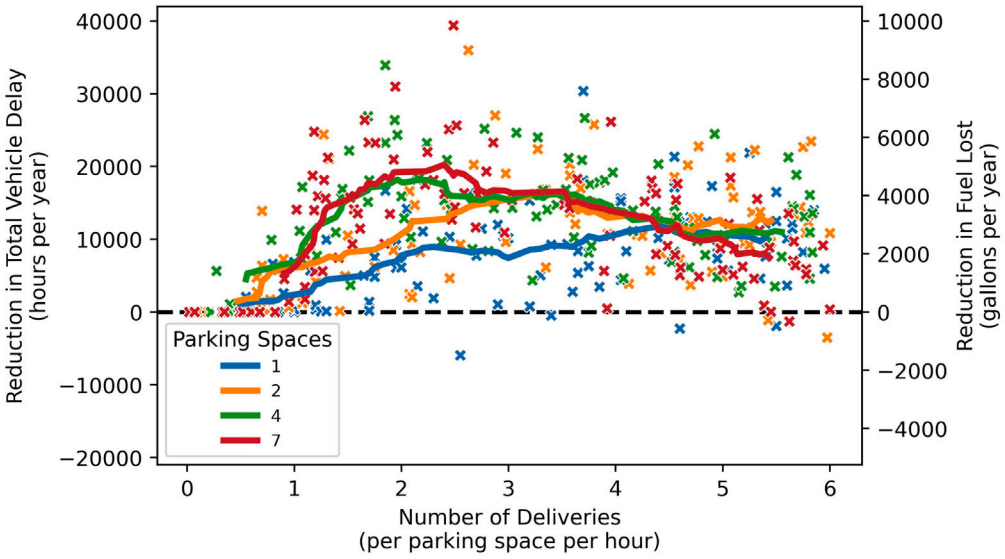
Fig. 10. A comparison of the reduction in double parking time at a set of co-located parking spaces per hour per parking space given ±5 min of scheduled arrival time flexibility with and without $\psi = 5$ min of buffer and random arrival time deviations drawn from $\mathcal{N}(0, \sigma^2)$ with $\sigma = 2$. Results from both the Aspen and Pittsburgh datasets are shown. Adding a buffer removes parking opportunities for delivery vehicles in the optimal schedule and therefore results in a lower reduction in double parking compared to the zero buffer scenario. In some instances, the optimal buffer schedule may result in more double parking than an uncoordinated model, as indicated by the negative reduction values in Fig. 10(b), shown below the dashed line. All graphics contain trend lines based on an 11-sample rolling average.

In an ideal case where delivery vehicles arrive to a set of co-located parking spaces precisely at their reservation start time, our analysis suggests that smart curb reservation systems reduce double parking across all scenarios examined, with greater potential when drivers have more arrival time flexibility and diminishing returns when more parking spaces are involved. In some cases smart curbspace can be more effective at reducing double parking than additional parking capacity (Fig. 9).

In cases where arrival times are uncertain and buffers are required between reservations, competitiveness drops, and reservation systems can increase double parking in some cases, such as when demand approaches saturation levels (Fig. 10(b)), or when arrival time uncertainty is high (Appendix Figure F.1b). With moderate arrival time uncertainty (8 min arrival uncertainty window) and buffers (5 min between reservations) and shorter parking duration times, we estimate that optimized, co-located reservation systems may save up to 12,000 h of traffic delay and 3000 gallons of gasoline annually (in our Pittsburgh case), whereas with large arrival time uncertainty (28 min arrival uncertainty window) and buffers (15 min between reservations) and long parking duration times, we estimate that optimized reservation systems may increase traffic delay by 4000 h and increase fuel consumption by 1000 gallons annually (in our Aspen case). These estimates imply that converting a single space to a reservation system could affect annual energy



(a) **Worst Case: Aspen data with 15 minute buffers**
Arrival uncertainty distributed $N(0,7)$



(b) **Best Case: Pittsburgh data with 5 minute buffers**
Arrival uncertainty distributed $N(0,2)$

Fig. 11. Reduction in estimated annual vehicle-hours of traffic delay and gallons of fuel consumed for optimally scheduled delivery vehicle arrivals relative to uncoordinated arrivals at a set of co-located parking spaces. Subplots show the worst (a) and best (b) case across scenarios with $\pm(\phi = 5)$ min of scheduled arrival flexibility, $\psi = 5$ or 15 min of buffer between reservations, $\sigma = 2$ or 7 standard deviation of uncertain vehicle arrival times, our base case traffic scenario, and demand based on the Aspen or Pittsburgh dataset. The ×'s show individual problem instances, and the curves show a 21-sample rolling average.

consumption by −100GJ to 400GJ, CO_2 emissions by −10 to 30 metric tons and externality costs by −$100,000 to +$300,000,[6] depending on operating conditions.

[6] Total annual minutes of vehicle delay scaled by our cost estimate for vehicle delay of $23/h (derived from the Department of Transportation (2016) and adjusted for 2021 median household income of $70,000. Also conservatively assumes only a single passenger is being delayed per vehicle).

Overall, our results suggest that optimized curb reservation systems can reduce double parking externalities in cases with low to moderate arrival time uncertainty, substantial demand, and relatively short duration requests, where missed reservations can frequently be filled in by other short duration stops. But in cases with high arrival time uncertainty and long duration reservations, reservation systems can increase double parking externalities, so such systems should be targeted to applications where they are likely to improve outcomes.

5.2. Limitations

Our model focuses on day-ahead reservation requests and we assume that delivery companies will adhere to the generated schedule as closely as possible. As a result, this does not allow for real-time parking requests or dynamic updates to arrival time during the delivery day. Instead, we use arrival time uncertainty to capture effects of deviation from a schedule planned in advance. We also add a buffer period between reservations to provision for the uncertainty in arrival times but recognize that this may not absorb all of the arrival uncertainty in a real-world scenario. As a result, our study results are likely most applicable to delivery vehicles that have arrival uncertainty within our buffer values, have a fewer number of stops on their route, as shown in Chiara et al. (2021) or have fewer stops that need a reservation and the remainder of their stops have flexibility in arrival.

In other applications, reservation requests may be made real-time in route to a delivery location, requiring a dynamic reservation system that can juggle parking assignments as new requests come in. We avoid such cases here because of our focus on comparing optimal results across scenarios. In dynamic reservation scenarios, system optimality is lost, and performance will depend on the details of dynamic implementation, such as how far in advance a driver needs to know whether or not they will be guaranteed a space or how frequently the algorithm updates. Future work on dynamic reservation systems could help to identify which such factors are important for making dynamic reservation systems competitive with the status quo.

Results of our study are based on parking data from Aspen, CO and Pittsburgh, PA and could vary in other situations. Our Aspen dataset lacks data on double-parked vehicles and contains many long duration, legally parked delivery vehicles. When we adopt these delivery patterns as demand requests in our simulations, some of these long duration events may double park when parking capacity is insufficient. In practice, double parked vehicles tend to park for shorter durations than legally parked vehicles Chiara et al. (2021), Schmid et al. (2018). Reservable curb benefits may also differ for locations with many short duration deliveries, such as parcel delivery vehicles in some cases Chiara et al. (2021), Jaller et al. (2021). The Pittsburgh dataset includes shorter duration delivery vehicles, but we cannot distinguish delivery vehicles from other trucks.

We assume all delivery vehicles unable to legally park will double park. In practice, some portion of such vehicles may choose to cruise in search of alternative parking or re-route to another destination, with different effects on congestion and energy Chiara et al. (2021), Jaller et al. (2021). Interactions with other parking in the vicinity but beyond the scope of our analysis could also have effects beyond what we model, and we do not model changes to costs and energy use imposed on delivery vehicles when shifting their scheduled arrival time. For example, vehicles that idle while delaying departure from a previous delivery location to accommodate a delayed reservation arrival time may induce other congestion, energy, or parking availability effects beyond our modeling scope. Additionally, we model a single location of co-located parking spaces recognizing that in a real-world application there may be many geographically disperse reservable spaces. Adding a geographic component adds realism and allows for parking schedules that encompass of a vehicle's full delivery tour.

Our model reflects delivery vehicle parking in a business district and does not include considerations for residential neighborhoods that have different challenges and configurations Chen et al. (2017), and our estimates of congestion and energy effects can be sensitive to assumptions about lane configuration and maximum queue size. Further, we assume that each parking space can support any type of delivery vehicle; if some vehicles require more space than others, this could affect results and model complexity.

Alternative objectives, such as minimizing lane obstruction or considering cost and revenue for the city, might produce different results, as might a range of implementation factors, including dynamic pricing of parking reservations to balance supply and demand, dynamic re-optimization with new requests and observations, enforcement Alho et al. (2018), Coord (2021b), Diehl et al. (2021), Muñuzuri et al. (2012), Nourinejad et al. (2020), accessibility and convenience of reservation systems (which could potentially make automatic reservations as part of a routing task), willingness to participate, and the ability of drivers to adjust arrival times.

CRediT authorship contribution statement

Aaron J. Burns: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Jeremy J. Michalek:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Constantine Samaras:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Constantine Samaras reports financial support was provided by US Department of Energy Vehicle Technologies Office. Jeremy J. Michalek reports financial support was provided by Mobility21 National University Transportation Center and the US Department of Energy Vehicle Technologies Office. Aaron J. Burns reports financial support was provided by Veteran Affairs, G.I. Bill.

Data availability

The code associated with this research is available at: https://github.com/burnsaar/Parking_Slot_Assignment_Problem-Public.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors of this paper did not use generative AI or any AI-assisted technologies in the writing of this manuscript.

Acknowledgments

This work was supported in part by the US Department of Energy's Vehicle Technologies Office, Award Number DE-EE0008463 and in part by Carnegie Mellon University's Mobility21 National University Transportation Center Federal Grant #69A3551747111, which is sponsored by the US Department of Transportation. Aaron Burns was also funded, in part, by a G.I. bill. The opinions and views expressed in this article are the authors' own and do not reflect the view of the United States government or any other organization. This research was initiated while C.S. was affiliated with Carnegie Mellon University. The views expressed in this paper do not reflect the views of the U.S. Government or any other organization. We are grateful to Professor Antonio Comi for helpful discussions on the implementation of his optimization algorithm; Professor Mireia Roca-Riu for insights on the MILP and ILP problem formulations and possible sources for delivery vehicle arrival datasets; Professor Markos Kladeftiras for discussions on details on the physical length of the simulated parking area in his double parking research; and Professor Sean Qian and Professor Chris Hendrickson for direction on our queuing model and help developing relevant parameters. Finally, this research would not have been possible without the continual support of family and friends including members of the Engineering and Public Policy and Mechanical Engineering Departments at Carnegie Mellon University.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.tre.2024.103574.

References

Abidi, S., Krichen, S., Alba, E., Molina, J.M., 2015. A new heuristic for solving the parking assignment problem. In: Procedia Computer Science, vol. 60, Elsevier B.V., pp. 312–321.
Alho, A.R., de Abreu e Silva, J., de Sousa, J.P., Blanco, E., 2018. Improving mobility by optimizing the number, location and usage of loading/unloading bays for urban freight vehicles. Transp. Res. D 61, 3–18.
Amaya, J., Encarnación, T., Delgado-Lindeman, M., 2023. Understanding delivery drivers' parking preferences in urban freight operations. Transp. Res. A 176.
Ayala, D., Wolfson, O., Xu, B., Dasgupta, B., Lin, J., 2011. Parking slot assignment games. In: Proceedings of the 19th ACM SIGSPATIAL International Conference on Advances in Geographic Information. Association for Computing Machinery, pp. 299–308.
Bierwirth, C., Meisel, F., 2015. A follow-up survey of berth allocation and quay crane scheduling problems in container terminals. European J. Oper. Res. 244 (3), 675–689.
Castrellon, J.P., Sanchez-Diaz, I., 2024. Effects of freight curbside management on sustainable cities: Evidence and paths forward. Transp. Res. D 130, 104165.
Chen, Q., Conway, A., Cheng, J., 2017. Parking for residential delivery in New York city: Regulations and behavior. Transp. Policy 54, 53–60.
Chen, Z., Yin, Y., He, F., Lin, J.L., 2015. Parking reservation for managing downtown curbside parking. Transp. Res. Rec. 2498, 12–18.
Chiara, G.D., Krutein, K.F., Ranjbari, A., Goodchild, A., 2021. Understanding urban commercial vehicle driver behaviors and decision making. Transp. Res. Rec..
Chrysostomou, K., Petrou, A., Aifadopoulou, G., Morfoulaki, M., 2019. Microsimulation modelling of the impacts of double-parking along an urban axis. In: Data Analytics: Paving the Way to Sustainable Urban Mobility (Ed.), Conference on Sustainable Urban Mobility. Springer, Cham, pp. 164–171.
Colorado Department of Transportation, 2020. Traffic data explorer.
Comi, A., Schiraldi, M.M., Buttarazzi, B., 2018. Smart urban freight transport: tools for planning and optimising delivery operations. Simul. Model. Pract. Theory 88, 48–61.
Coord, 2021a. Curbs, COVID and Busways a Case Study of Brooklyn's Fulton Mall. Technical Report.
Coord, 2021b. Smart Zones Deliver Aspen Smart Zone Pilot Program. Technical Report.
Delaitre, L., 2009. A new approach to diagnose urban delivery plans. In: International Conference on Computers & Industrial Engineering. IEEE.
Department of Transportation, 2016. The Value of Travel Time Savings: Departmental Guidance for Conducting Economic Evaluations Revision 2 (2016 Update). Technical Report.
Desaulniers, G., Madsen, O.B.G., Ropke, S., Toth, P., Vigo, D., 2015. The vehicle routing problem with time windows. In: Toth, P., Vigo, D. (Eds.), Vehicle Routing Problems, Methods, and Applications, second ed. Society for Industrial and Applied Mathematics, (Chapter 5).
Dey, S.S., Pérez, B.O., Richards, L., Pochowski, A., Sanders, M., Darst, M., Sanchez, E.C., Dock, S., 2019. Demystifying urban curbside freight management: Strategic incremental approach from Washington, D.C. Transp. Res. Rec. 2673 (12), 312–326.
Diehl, C., Ranjbari, A., Goodchild, A., 2021. Curbspace management challenges and opportunities from public and private sector perspectives. Transp. Res. Rec..
Energy.gov, 2015. Idle fuel consumption for selected gasoline and diesel vehicles.
Errousso, H., El Ouadi, J., Abdellaoui Alaoui, E.A., Benhadou, S., 2021. Dynamic parking space allocation at urban scale: Problem formulation and resolution. J. King Saud Univ. Comput. Inf. Sci..
Girón-Valderrama, G.d.C., Machado-León, J.L., Goodchild, A., 2019. Commercial vehicle parking in downtown seattle: Insights on the battle for the curb. Transp. Res. Rec. 2673 (10), 770–780.
Golias, M., Assistant, G.R., Boile, M., Theofanis, S., 2007. The berth allocation problem: A formulation reflecting time window service deadlines. In: 48th Transportation Research Forum Annual Meeting.
Hakeem, A., Gehani, N., Ding, X., Curtmola, R., Borcea, C., 2016. On-The-Fly Curbside Parking Assignment. Technical Report, New Jersey Institute of Technology.
Huang, Y.H., Hsieh, C.H., 2022. A decision support system for available parking slots on the roadsides in urban areas. Expert Syst. Appl. 205.
Irnich, S., Toth, P., Vigo, D., 2014. The family of vehicle routing problems. In: Toth, P., Vigo, D. (Eds.), Vehicle Routing Problems, Methods, and Applications, second ed. Society for Industrial and Applied Mathematics, (Chapter 1).

Iwan, S., Kijewska, K., Johansen, B.r.G., Eidhammer, O., Małecki, K., Konicki, W., Thompson, R.G., 2018. Analysis of the environmental impacts of unloading bays based on cellular automata simulation. Transp. Res. D 61, 104–117.
Jaller, M., Holguín-Veras, J., Hodge, S., 2013. Parking in the city: Challenges for freight traffic. Transp. Res. Rec. (2379), 46–56.
Jaller, M., Rodier, C., Zhang, M., 2021. Fighting for Curb Space: Parking, Ride-Hailing, Urban Freight Deliveries, and Other Users. Technical Report.
Jaller, M., Xiao, R.I., Dennis-Bauer, S., Rivera-Royero, D., Pahwa, A., 2023. Estimating last-mile deliveries and shopping travel emissions by 2050. Transp. Res. D 123.
Jiang, B., Fan, Z.P., 2020. Optimal allocation of shared parking slots considering parking unpunctuality under a platform-based management approach. Transp. Res. E 142.
Johansen, B.G., Andersen, J., Levin, T., 2014. Better use of delivery spaces in Oslo. Procedia Soc. Behav. Sci. 151, 112–121.
Kim, W., Wang, X., 2021. Double parking in New York city: a comparison between commercial vehicles and passenger vehicles. Transportation.
Kladeftiras, M., Antoniou, C., 2013. Simulation-based assessment of double-parking impacts on traffic and environmental conditions. Transp. Res. Rec. 2390 (1), 121–130.
Letnik, T., Farina, A., Mencinger, M., Lupi, M., Božičnik, S., 2018. Dynamic management of loading bays for energy efficient urban freight deliveries. Energy 159, 916–928.
Liu, J., Ma, W., Qian, S., 2023. Optimal curbside pricing for managing ride-hailing pick-ups and drop-offs. Transp. Res. C 146.
Mannering, F., Washburn, S., 2020. In: Brady, J. (Ed.), Principles of Highway Engineering and Traffic Analysis, seventh ed. Wiley & Sons, Hoboken, NJ.
Mejri, N., Ayari, M., Langar, R., Saidane, L., 2016. Reservation-based multi-objective smart parking approach for smart cities. In: IEEE 2nd International Smart Cities Conference: Improving the Citizens Quality of Life, ISC2 2016 - Proceedings. Institute of Electrical and Electronics Engineers Inc..
Mladenović, M., Delot, T., Laporte, G., Wilbaut, C., 2021. A scalable dynamic parking allocation framework. Comput. Oper. Res. 125.
Mor, A., Speranza, M.G., Viegas, J.M., 2020. Efficient loading and unloading operations via a booking system. Transp. Res. E 141.
Muñuzuri, J., Cortés, P., Guadix, J., Onieva, L., 2012. City logistics in Spain: Why it might never work. Cities 29 (2), 133–141.
NADA Guides, 2021. How long is the average car?.
Nourinejad, M., Gandomi, A., Roorda, M.J., 2020. Illegal parking and optimal enforcement policies with search friction. Transp. Res. E 141.
Onufer, M., 2020. DOMI Awarded $100,000 Grant to Implement Curbside Parking Pilot. Technical Report, City of Pittsburgh, Pittsburgh.
Park, Y.-M., Kim, K.H., 2003. A scheduling method for berth and quay cranes. OR Spectrum 25, 1–23.
Parry, I.W.H., Walls, M., Harrington, W., 2007. Automobile Externalities and Policies. Technical Report, pp. 373–399.
PennDOT, 2021. Traffic volume maps.
Peterson, S.S., 2021. Pittsburgh pilots new smart loading zones to ease congestion and air pollution from delivery vehicles.
Pilz, D., Schwerdfeger, S., Boysen, N., 2022. Make or break: Coordinated assignment of parking space for breaks and rest periods in long-haul trucking. Transp. Res. B 164, 45–64.
Portilla, A.I., Borja, Oreña, A., Luis, J., Berodia, M., José, F., Díaz, R., 2009. Using M / M / queueing model in on-street parking maneuvers. J. Transp. Eng. 135 (8).
Ramirez-Rios, D.G., Kalahasthi, L.K., Holguín-Veras, J., 2023. On-street parking for freight, services, and e-commerce traffic in US cities: A simulation model incorporating demand and duration. Transp. Res. A 169.
Roca-Riu, M., Fernández, E., Estrada, M., 2015. Parking slot assignment for urban distribution: Models and formulations. Omega (UK) 57, 157–175.
Sayarshad, H.R., Sattar, S., Oliver Gao, H., 2020. A scalable non-myopic atomic game for a smart parking mechanism. Transp. Res. E 140.
Schmid, J., Wang, X.C., Conway, A., 2018. Commercial vehicle parking duration in New York city and its implications for planning. Transp. Res. A 116, 580–590.
Schrank, D., Albert, L., Eisele, B., Lomax, T., 2021. 2021 Urban Mobility Report. Technical Report, Texas A&M.
Schrank, D., Eisele, B., Lomax, T., 2019. 2019 Urban Mobility Report. Technical Report, Texas A&M.
Simoni, M.D., Claudel, C.G., 2018. A simulation framework for modeling urban freight operations impacts on traffic networks. Simul. Model. Pract. Theory 86, 36–54.
Spadafora, J., Rodriguez, M., 2021. Pitney bowes parcel shipping index reveals 37 percent parcel volume growth in US for 2020.
TheColdWire.com, 2021. How long is a block? (everything to know).
Transportation Research Board Committee on Highway Capacity and Quality of Service, 2000. Highway Capacity Manual. National Academy of Sciences.
Yang, K., Roca-Riu, M., Menéndez, M., 2019. An auction-based approach for prebooked urban logistics facilities. Omega (UK) 89, 193–211.
Zhang, L., Ding, P., Thompson, R.G., 2023. A stochastic formulation of the two-echelon vehicle routing and loading bay reservation problem. Transp. Res. E 177, 103252.